

A CPI-Weighted Web-Based Price Index for Monitoring Inflation and Cost of Living in Türkiye

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Online Prices as a Transparent Complement to Official Inflation

Daily, reproducible measurement



High price volatility

Food, rent and household prices move fast and hit purchasing power directly.



The reproducibility gap

Official CPI is the standard reference, but its collection and aggregation cannot be reproduced from outside.



The digital alternative

Daily crawling keeps basket logic while adding high-frequency visibility.

What this study delivers

Open dataset

34.2M price records and 568,204 unique items, seven consumer domains.

Reproducible pipeline

Crawl, clean, match, normalize. Code and data published.

Two measures

A CPI-weighted web index, and a nutrition-basket food cost estimate.

Six Benchmarks, and the Gap This Study Fills

Where a daily multi-category web index sits among existing price indicators.

01 Official CPI TUIK · Monthly	02 Alternative CPI ENAG · High frequency	03 Regional index ITO · Istanbul
04 Web scraping research Eurostat · ONS · BPP	05 Food affordability TÜRK-İŞ · Monthly	06 This study Daily · 7 domains

Closest predecessor, and what this study adds

Soybilgen, Yazgan & Kaya (2023)

Daily online prices for Türkiye, but food subindexes only. One COICOP group.

This study

Seven consumer domains, official COICOP weights, published collection and calculation code.

34.2M Price Records, Seven Domains, March to May 2026

62.04% of the official TUIK basket weight, collected from seven consumer domains.

34.2M

Price records

568,204

Unique items

7 + 1

Domains, health aux.

44 / 60+

Stores / cities

Unique items and raw rows by category (Table I)

Category	Unique items	Raw rows
Clothing	140,297	7.77M
Supermarkets	128,527	7.39M
Construction	104,936	7.44M
Cosmetics	88,908	2.15M
Home goods	66,113	1.87M
Technology	26,362	0.54M
Rental housing	13,061	7.05M
Total	568,204	34.2M

Geographic and basket coverage



Collection and cleaning

Matching

Product ID or normalized name

Dropped

Duplicates, missing and broken prices

Outlier Rule

Moves over 80% count as parsing errors

No imputation

Missing at either endpoint leaves the window

CPI-Weighted Price Index Aggregation Pipeline

Standardized processing pipeline ensuring endpoint-matching integrity, outlier filtering, and COICOP weight re-normalization.

STAGE 01

Pair Matching

Inner-join snapshots by store, product name, COICOP, & sector for common products.

STAGE 02

Store Averaging

Collapse duplicate store rows and compute equal-weighted cross-store changes.

STAGE 03

Outlier Removal

Exclude product pairs where absolute price changes exceed the 80% threshold.

STAGE 04

CPI Aggregation

Re-normalize TUIK expenditure weights across valid observed categories to 100%.

📊 Laspeyres-Type Aggregation Formulas

$$r_{j,s,t} = 100 \cdot \frac{p_{j,s,t}}{p_{j,s,t-1}} - 1$$
$$\pi_t = \sum_{c \in C_t} w_{c,t} \bar{r}_{c,t}, \quad w_{c,t} = \frac{w_c}{\sum_{k \in C_t} w_k}$$

☰ Quality Control & Aggregation Protocols

- ✔ **Equal Store Weighting:** When identical products appear across multiple retail chains, store-level changes are averaged with equal weight before category rollout.
- ✔ **Outlier Filtering ($|\Delta P| > 80\%$):** Price variations exceeding 80% between snapshots are automatically excluded to eliminate scraping syntax or packaging artifacts.
- ✔ **Independent Monthly Windows:** Each monthly rate is computed directly from two endpoint snapshots without chaining to maintain an uncorrupted panel basis.

Nutrition-Based Hunger Threshold Estimation Framework

Dietitian-supported 16-item food basket tracking minimum food expenditure for a four-person household across 13 supermarkets.

STAGE 01

Basket Design

16 fixed food items aligned with Turkish Dietary Guidelines (TUBER 2022) nutrition norms.

STAGE 02

Price Crawling

Extract raw food prices across 13 major Turkish online supermarket chains.

STAGE 03

Store Aggregation

Standardize units and summarize items by mean, median, and minimum prices.

STAGE 04

Threshold Evaluation

Compute total monthly basket cost; apply 30.70% ratio for poverty threshold benchmark.

 Household Basket Cost Formulation

$$H_t^{(a)} = \sum_{i=1}^{16} q_i \cdot p_{i,t}^{(a)}$$

$a \in \{ \text{mean}, \text{median}, \text{minimum} \}$

 Nutrition Guidelines & Aggregation Rules

- ✔ **Fixed Quantity Invariance:** Quantities are kept perfectly constant across months to ensure cost changes solely isolate pure food price inflation.
- ✔ **Tri-Metric Strategy:** Mean captures market exposure, median filters price extremes, and minimum defines an optimal lower-bound consumer benchmark.

Rule-Based Matching & Unit Normalization Engine

Deterministic transformation pipeline reconciling disparate online packaging units, brand tokens, and overlapping sector taxonomies.

STAGE 01

Text Sanitization

Lowercase conversion, special character removal, and whitespace standardization.

STAGE 02

Unit Extraction

Regex parsing for expressions ('1 LT', '500 GR', '1000 ML) to common metric base.

STAGE 03

Unit Price Conversion

Scale package prices to metric standards (TRY/kg, TRY/liter, TRY/piece).

STAGE 04

Deduplication

Average overlapping items across retail sectors (e.g. Group 05 Home & Construction).

↶ Unit Price Normalization Schema (Fig. 1)



$$p_i^{std} = p_i \cdot \frac{1}{u_i}$$

⚙️ Matching & Normalization Protocols

- ✓ **Rule-Based Token Matching:** Scraped products match basket slots using strict keyword validation, exclusion filters, and volume tokens to eliminate mismatches.
- ✓ **Domain-Specific Pricing:** Food products use standardized unit prices (TRY/kg or L), while non-food (clothing, tech) rely on cleaned nominal prices.
- ✓ **Zero Imputation Guarantee:** Products that cannot be matched unambiguously across endpoints are dropped to prevent synthetic price distortion.

Inflation Estimates and Category Analysis

Table 3. Reference Source Comparison (%)

SOURCE	MAR	APR	MAY
TUIK (official CPI)	1.94	4.18	1.71
ENAG (independent)	4.10	5.07	2.16
ITO (Istanbul)	2.97	3.74	1.53
This study	2.00	1.16	0.62

Table 6. Category-Level Inflation (%)

CATEGORY	CO.	MAR	APR	MAY
Food (supermarkets)	01	2.22	1.84	1.26
Clothing	03	-	-1.34	-0.42
Home goods & const.	05	-	1.95	0.42
Cosmetics	06	-	-	-0.69
Rental housing	04	-1.59	1.62	2.06
Aggregate (TUIK-w)	-	2.00	1.16	0.62

NOTES & METHODOLOGY

Red: No reliable paired data. **Cyan:** Rental housing reported separately and excluded from aggregate. **d** Values are rounded arithmetic means.

Monthly Food Basket (Four-Person Household)

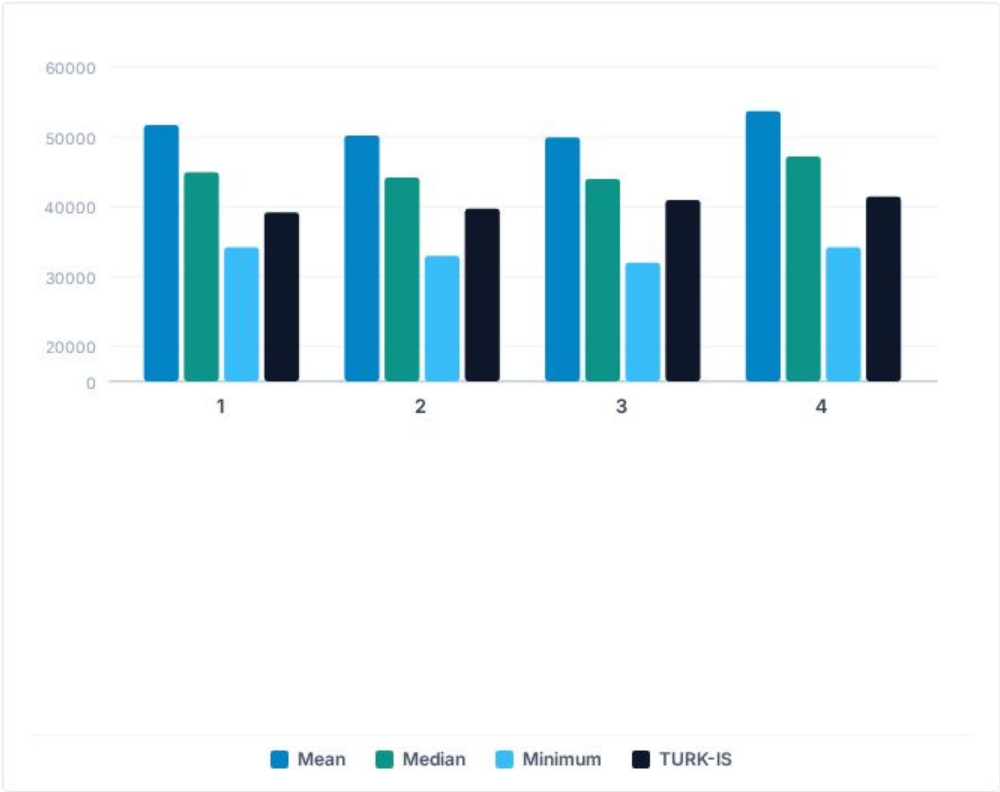
Prepared with dietitian support, aligned with TUBER 2022 · quantities in kg unless stated

Dairy Products Yogurt 59.7 · White cheese 5.7	Fruits Banana 16.7 · Seasonal fruit 12.9
Meat and Protein Cubed/lamb meat 4.6 · Chicken 10.3 · Fish 6.9 · Eggs 120 pcs	Vegetables Onion 18.0 · Eggplant/Zucchini 23.1 · Other vegetables 11.8
Legumes Chickpeas 1.8	Oils Olive oil 1.1 liter
Nuts and Seeds Walnut / Hazelnut / Peanut 2.7	Other Foods Grissini 2.1
Grains Bread 18.0	

Reference: Study \$IV-B & Table VII; TUBER 2022

Hunger Threshold Estimates vs TURK-IS

Four-person household, monthly food expenditure (TRY) · February–May 2026



● **Mean**
34.9% – 51.3% above TURK-IS

● **Median**
12.2% – 23.5% above TURK-IS

● **Minimum**
20.3% – 34.2% below TURK-IS

The minimum-price estimate is a lower bound: it assumes consumers can always buy the cheapest qualifying product.

Discussion & Open Source

Scan the QR code to access our GitHub repository for code and data replication

Access the Repository

Scan the QR code on the right with your mobile device to view our complete codebase, data cleaning pipelines, and replication scripts.

- [Open Source License](#)
- [Reproducible Notebooks](#)
- [Raw Web-Scraped Datasets](#)

